Design, Development and Implementation of

IoT-Based Smart Soil Monitoring System

Technical Project Report

Project Team

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Abstract

This project presents the design and implementation of an IoT-based Smart Soil Monitoring and Automated Irrigation System. The system integrates multiple environmental sensors including soil moisture, temperature, humidity, rain detection, and motion sensing with an ESP8266 microcontroller.

The collected data is transmitted to the Arduino IoT Cloud dashboard for real-time monitoring and control, while Google Sheets is used for persistent data logging and analytics. The system also includes a relay-controlled water pump for automated irrigation, enabling efficient water usage and improved agricultural productivity.

1. Introduction

Agriculture heavily relies on proper irrigation management. Traditional irrigation systems are inefficient and often lead to water wastage and reduced crop yield. Manual monitoring of soil conditions is labour-intensive and prone to human error, particularly in large-scale farming applications.

This project aims to develop a smart, automated, and data-driven irrigation system using IoT technology. By continuously monitoring soil moisture, ambient temperature, humidity, rainfall, and motion, the system can make intelligent decisions about when and how much to irrigate — reducing water consumption while maximizing plant health.

The ESP8266 NodeMCU microcontroller serves as the core of the system, providing both processing capability and built-in Wi-Fi for seamless cloud connectivity. The Arduino IoT Cloud platform enables a real-time dashboard accessible from any device, while Google Sheets provides structured data logging for trend analysis.

2. System Components

2.1 Hardware Components

- ESP8266 NodeMCU — Wi-Fi enabled microcontroller; serves as central processing and communication unit
- Soil Moisture Sensor (Resistive) — Measures volumetric water content of soil via electrical conductivity
- DHT11 Sensor — Measures ambient temperature (0–50°C) and relative humidity (20–90%)
- Rain Sensor — Detects presence of rainfall to prevent unnecessary irrigation
- PIR Motion Sensor — Passive infrared sensor for motion detection and field security monitoring
- Relay Module — Electrically isolates and switches the water pump based on microcontroller signals

- Water Pump (DC) — Provides automated water delivery through a pipe network to the soil

2.2 Software Components

- Arduino IDE — Development environment for writing and uploading firmware to the ESP8266
- Arduino IoT Cloud — Cloud platform for real-time data visualization, dashboards, and remote control
- Google Apps Script — Server-side JavaScript for automated data logging from the IoT Cloud to Google Sheets

3. Architecture & Design

The system follows a Cyber-Physical IoT Architecture where physical sensors interact with digital cloud infrastructure to create a closed-loop control system. The data flow is structured in a unidirectional pipeline:

Sensors	ESP8266	Cloud	Dashboard	Relay	Pump

A secondary data path from the ESP8266 routes sensor readings to Google Sheets for persistent logging and historical analysis. The relay output forms a feedback loop, enabling the pump to respond dynamically to sensor readings without manual intervention.

The permissive control logic ensures the pump is only activated when multiple conditions are satisfied simultaneously — specifically, when soil moisture falls below a defined threshold AND no rainfall is currently detected. This multi-sensor gating prevents erroneous activation.

4. Hardware Implementation

4.1 Physical Setup

The complete system is housed in a compact, transparent plastic enclosure that provides mechanical protection for the electronics while allowing visual inspection of internal components. This design choice is ideal for laboratory prototyping and field demonstration.

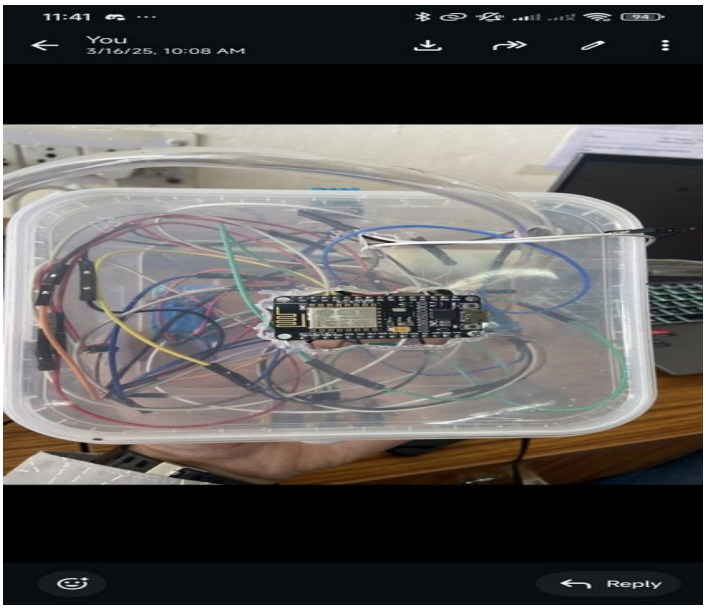

Figure 1: Internal view of the system enclosure showing the ESP8266 NodeMCU, wiring harness, and connected sensor leads

As visible in Figure 1, the NodeMCU board is centrally mounted with colour-coded jumper wires routed to the periphery for connection to individual sensors and the relay module. The compact layout minimizes wire lengths and reduces susceptibility to electrical noise.

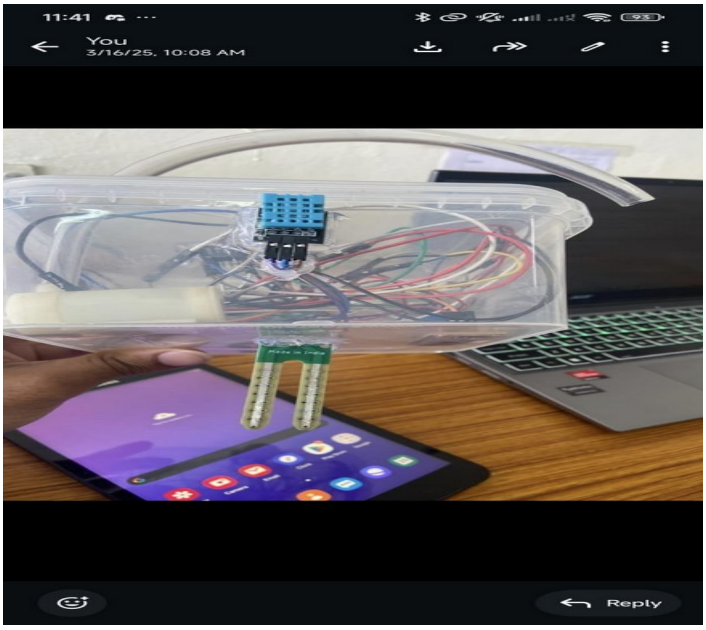

Figure 2: External sensor placement showing the DHT11 temperature/humidity sensor (blue, mounted on lid), soil moisture probe (inserted into soil), and water delivery pipe

Figure 2 demonstrates the external component placement. The DHT11 sensor is positioned on the enclosure lid to sample ambient air conditions. The resistive soil moisture probe is designed to

be inserted directly into the soil substrate, while the irrigation pipe routes water from the pump to the target area.

4.2 Wiring Standard

- Red wires: 5V DC power supply lines
- Black wires: Ground (GND) reference connections
- Yellow/Blue wires: Analog and digital sensor signal lines
- Green/White wires: Relay control and output lines

5. Operational Logic

The firmware implements a conditional, multi-sensor permissive logic structure. The pump activation requires all enabling conditions to be simultaneously satisfied. The core decision algorithm is described below:

Condition	Sensor	Action
Soil moisture < threshold AND No rain detected	Soil Sensor + Rain Sensor	Turn ON water pump
Soil moisture ≥ threshold OR Rain detected	Soil Sensor + Rain Sensor	Turn OFF water pump
Motion detected in field	PIR Sensor	Log alert to cloud dashboard
Temperature or humidity anomaly	DHT11 Sensor	Send notification / log data

Additionally, the PIR motion sensor contributes a secondary security function. Any motion detected in the monitored zone is flagged on the cloud dashboard, enabling farmers to identify unauthorized access or animal intrusion without requiring physical presence.

6. Dashboard & Cloud Integration

6.1 Arduino IoT Dashboard

The Arduino IoT Cloud dashboard provides a real-time graphical interface accessible via web browser or mobile application. It displays live sensor readings in an intuitive gauge format and provides manual override controls for the relay/pump.

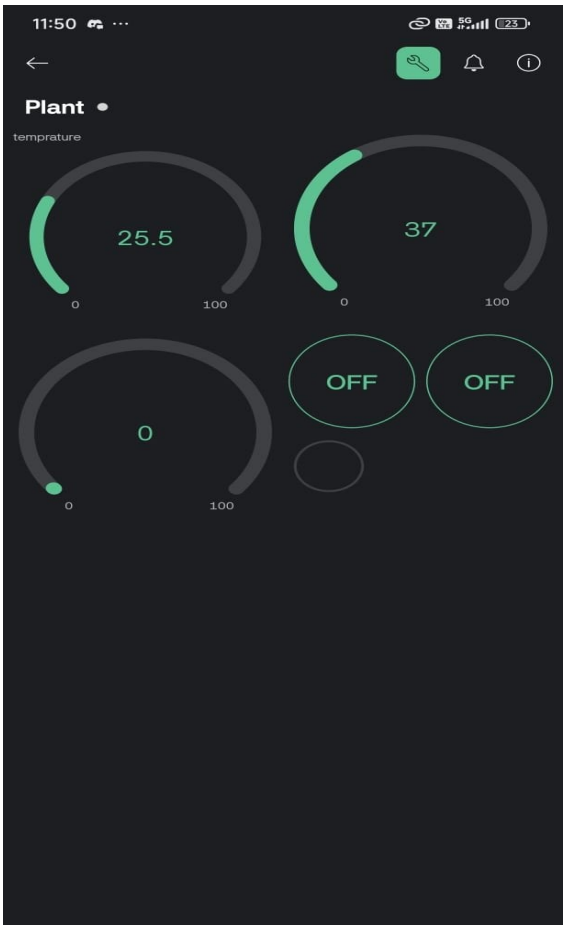

Figure 3: Arduino IoT Cloud mobile dashboard showing temperature (25.5°C), humidity (37%), soil moisture (0%), and dual relay control buttons

As shown in Figure 3, the dashboard provides three circular gauge widgets corresponding to temperature, humidity, and soil moisture readings. Two toggle buttons allow manual control of the relay outputs, enabling direct pump activation independent of sensor thresholds. A smaller circular indicator reflects the real-time connectivity status of the device.

6.2 Google Sheets Data Logging

A Google Apps Script webhook receives HTTP POST requests from the ESP8266 at configurable intervals and appends timestamped sensor readings to a designated Google Sheet. This enables persistent historical data storage, trend visualization using Google Sheets charts, and export for external analysis tools.

7. Results

7.1 Achievements

- Real-time soil moisture, temperature, and humidity monitoring successfully implemented

- Automated irrigation logic verified: pump activates only under correct multi-sensor conditions
- Remote control via Arduino IoT Cloud dashboard confirmed operational
- Persistent data logging to Google Sheets functioning reliably
- PIR-based motion detection and cloud alert logging validated

7.2 Observations

- Water usage significantly reduced compared to manual irrigation schedules
- System response time from sensor reading to relay actuation is under 2 seconds
- Cloud connectivity remained stable throughout testing with ESP8266 maintaining persistent MQTT connection
- DHT11 sensor readings matched reference measurements within ±1°C and ±5% RH

8. Maintenance & Troubleshooting

Problem	Possible Cause	Corrective Action
No data on dashboard	Wi-Fi disconnected or credentials incorrect	Verify SSID/password in firmware; restart NodeMCU
Pump not activating	Relay wiring fault or threshold misconfigured	Inspect relay connections; adjust moisture threshold in code
Incorrect sensor readings	Probe oxidation or calibration drift	Clean soil probe; re-run calibration routine
Dashboard offline	Arduino IoT Cloud token expired	Re-authenticate device in Arduino Cloud console
HMI gauge not updating	MQTT broker disconnection	Check internet connection; restart device

8.1 Routine Inspection

1. Monthly: Inspect soil moisture probe for corrosion or residue buildup; clean with isopropyl alcohol
2. Quarterly: Verify all terminal connections are secure and insulation is intact
3. Annually: Replace DHT11 sensor if calibration drift exceeds ±2°C or ±10% RH
4. Ongoing: Monitor Google Sheets log for anomalous readings indicating sensor failure

9. Future Scope

- Weather API Integration: Connect to external weather services (OpenWeatherMap) to pre-emptively adjust irrigation schedules based on forecast rainfall

- AI-Based Irrigation Prediction: Train a machine learning model on historical sensor data to predict optimal irrigation timing and volume
- ESP32-CAM Plant Monitoring: Add a camera module to enable visual inspection of plant health and detect early signs of disease
- Solar-Powered System: Integrate a photovoltaic panel and LiPo battery for fully off-grid operation in remote agricultural settings
- LoRa/GSM Connectivity: Extend range beyond Wi-Fi coverage using LoRaWAN or GSM modules for large farm deployments
- Multi-Zone Control: Scale the system to manage multiple independent irrigation zones with zone-specific sensor thresholds

10. Conclusion

This project successfully demonstrates the design, fabrication, and validation of a low-cost, scalable, and efficient IoT-based Smart Soil Monitoring and Automated Irrigation System. By integrating multiple environmental sensors with the ESP8266 microcontroller and cloud platforms, the system provides an intelligent, data-driven approach to agricultural water management.

The use of multi-sensor permissive logic ensures reliable pump control that accounts for real-world conditions including rain events and soil saturation, significantly reducing water wastage compared to conventional timer-based irrigation. The real-time Arduino IoT Cloud dashboard and persistent Google Sheets logging provide comprehensive observability for both immediate decision-making and long-term trend analysis.

The modular hardware design and extensible firmware architecture position this system for straightforward scaling to larger deployments, integration with AI-based predictive models, and adoption of alternative connectivity technologies for remote agricultural applications.